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Impact of Schizophrenia Spectrum Disorders on the Receipt of Invasive and Systemic Therapy for Colorectal Cancer: A Nationwide Multicenter Retrospective Cohort Study in Japan

Masaki Fujiwara[1] | Yuto Yamada[2,3] | Taisuke Ishii[4] | Tomone Watanabe[4] | Maiko Fujimori[5] | Naoki Nakaya[6] | Toshihiko Kawamura[7] | Koji Otsuki[8] | Kunitoshi Shigeyasu[9] | Taichi Shimazu[10] | Shiro Hinotsu[11] | Yosuke Uchitomi[12] | Masatoshi Inagaki[8]

[1]Department of Neuropsychiatry, Medical Development Field, Okayama University, Okayama, Japan | [2]Department of Neuropsychiatry, Okayama University Graduate School of Medicine, Dentistry, and Pharmaceutical Sciences, Okayama, Japan | [3]Department of Psychiatry, Sekizen Hospital, Tsuyama, Japan | [4]Division of Health Services Research, National Cancer Center Institute for Cancer Control, National Cancer Center, Tokyo, Japan | [5]Division of Survivorship Research, National Cancer Center Institute for Cancer Control, National Cancer Center, Tokyo, Japan | [6]Tohoku Medical Megabank Organization, Tohoku University, Sendai, Japan | [7]Department of Medical Informatics, Shimane University Hospital, Izumo, Japan | [8]Department of Psychiatry, Faculty of Medicine, Shimane University, Izumo, Japan | [9]Department of Gastroenterological Surgery, Medical Development Field, Okayama University, Okayama, Japan | [10]Division of Behavioral Sciences, National Cancer Center Institute for Cancer Control, National Cancer Center, Tokyo, Japan | [11]Department of Biostatistics and Data Management, Sapporo Medical University, Sapporo, Japan | [12]Department of Cancer Survivorship and Digital Medicine, The Jikei University School of Medicine, Tokyo, Japan

Correspondence: Masatoshi Inagaki (minagaki@med.shimane-u.ac.jp)

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ABSTRACT

Introduction: This study examined treatment disparities for colorectal cancer among patients diagnosed with schizophrenia spectrum disorders (SSD), focusing on invasive treatments and stage-appropriate systemic therapy within a universal healthcare system.

Method: In this nationwide retrospective cohort study (2018–2021), we identified 248,966 colorectal cancer patients, including 2337 diagnosed with SSD, using linked cancer registry and insurance claims data in Japan. The presence of SSD was classified according to ICD-10 codes F20–29. We used multivariable logistic regression to compare the odds of receiving stage-appropriate adjuvant chemotherapy and systemic therapy, as well as the odds of receiving surgical or endoscopic treatments, between the two groups. The analysis adjusted for age, sex, clinical stage, and scores on the Charlson Comorbidity Index and Barthel Index.

Results: The clinical stage distribution at diagnosis for colorectal cancer differed significantly between patients with SSD and those without psychiatric disorders ($p < 0.001$). After adjusting for clinical stage and other covariates, patients with SSD demonstrated significantly lower odds of receiving surgical or endoscopic treatment (adjusted odds ratio [aOR], 0.83; 95% CI, 0.73–0.94). The disparities were more pronounced for systemic therapy; patients with SSD had substantially lower odds of receiving adjuvant chemotherapy for stage III disease (aOR, 0.33; 95% CI, 0.26–0.41) and systemic therapy for stage IV disease (aOR, 0.23; 95% CI, 0.17–0.31).

Masaki Fujiwara and Yuto Yamada contributed equally to this work.

Conclusion: Patients with SSD encounter substantial disparities in accessing standard colorectal cancer care, particularly systemic therapies. These findings highlight the urgent need for interventions to ensure equitable cancer treatment for this vulnerable population.

1 | Introduction

1.1 | Background

Patients diagnosed with schizophrenia spectrum disorders (SSD) have an approximately 1.69-fold increased risk of mortality due to colorectal cancer compared with the general population [1]. However, the question of whether the incidence of colorectal cancer differs significantly between patients with SSD and the general population remains unresolved [2–7]. Therefore, the observed elevated mortality cannot be fully explained by differences in cancer incidence alone. Inequalities in access to and uptake of cancer screening and treatment services may contribute to this increased mortality, highlighting the urgent need to address these disparities [8–11].

Focusing on surgical treatment, disparities in colorectal cancer care for patients with severe mental illness (SMI), including SSD, have been reported across multiple countries [12–18]. A meta-analysis by Protani et al. [19] showed that patients with SMI are less likely to undergo surgery (RR = 0.90; 95% CI, 0.84–0.98; $I^2 = 76\%$; $p = 0.003$). A recent study in Sweden found that although patients with comorbid SMI were more likely to receive emergency surgery than patients with no history of mental illness, there was no significant difference in the rate of surgical resection for stage I–III colorectal cancer [18]. These findings suggest that the extent of disparities in surgical treatment for colorectal cancer among patients with SSD may vary among national healthcare systems. In addition, potential sex differences in the relationship between SSD and the receipt of invasive colorectal cancer treatment remain underexplored.

However, surgery is not the only treatment modality essential for improving the prognosis of patients with colorectal cancer. There is substantially less information about disparities in systemic therapy than about disparities in surgery. Only a few studies have specifically examined whether patients with SSD receive stage-appropriate systemic therapy for colorectal cancer. A few studies have reported disparities in the receipt of postoperative chemotherapy among patients with SMI [16–18]. Furthermore, although patients with stage IV colorectal cancer and SSD may encounter barriers to receiving systemic therapy, there is little evidence regarding whether they actually receive guideline-recommended systemic treatment. Because systemic therapies typically require prolonged adherence, frequent outpatient visits, and self-management of adverse effects, the disparities may be even greater than for surgery. This knowledge gap is particularly concerning given that recent advances in systemic therapy have led to significant improvements in survival outcomes for colorectal cancer [20].

1.2 | Aims of the Study

This study analyzed nationwide data from Japan to examine whether patients diagnosed with SSD are less likely than those with no diagnosed psychiatric disorder to receive stage-appropriate systemic therapy for colorectal cancer, as well as invasive treatments including surgery and endoscopic procedures.

2 | Methods

2.1 | Study Design

We conducted a multicenter retrospective cohort study using data curated by the Quality-of-care Monitoring Project, a nationwide research project conducted by the National Cancer Center Japan [21].

2.2 | Setting

The Japanese population, comprising approximately 120 million individuals, is served by a publicly funded universal health care system. This system provides coverage for all residents and aims to ensure equitable access to high-quality medical care. Additionally, anyone enrolled in public health insurance in Japan is eligible for the "High-cost Medical Expense Benefit," which limits monthly out-of-pocket expenditures to a predetermined maximum amount, payable at the point of service [22]

2.3 | Data Sources

The Quality-of-Care Monitoring Project integrates data from the Hospital-Based Cancer Registry (HBCR) national database with diagnosis procedure combination (DPC) data, which are equivalent to health insurance claims. The HBCR compiles records on cancer care provided at cancer hospitals designated by the Japanese Ministry of Health, Labour and Welfare, as well as from certain non-designated hospitals that participate voluntarily [23]. The data include detailed information on patient age at cancer diagnosis, cancer type, extent of disease, clinical and pathological cancer stage, classification based on the 8th edition of the tumor-node-metastasis classification, and tumor coding based on the International Classification of Diseases for Oncology, 3rd Edition (ICD-O-3). As of 2020, the Quality-of-care Monitoring Project accounted for approximately 65% of all new cancer diagnoses in Japan [24, 25].

The project collects DPC data for patients registered in the HBCR from October of the year preceding the cancer diagnosis until the end of March in the second year following diagnosis. In this study, DPC data were successfully linked to 94.8% of patients included in the HBCR.

2.4 | Participants

This study included patients who commenced treatment for colorectal cancer between January 1, 2018, and December 31, 2021,

Summary

- Significant outcomes
  - Patients diagnosed with schizophrenia spectrum disorders (SSD) were significantly less likely to receive surgical or endoscopic treatment for colorectal cancer compared with patients without psychiatric disorders.
  - Disparities were particularly pronounced for systemic therapy; patients with SSD had markedly lower odds of receiving guideline-recommended adjuvant chemotherapy (stage III) and systemic therapy (stage IV).
  - These findings highlight the urgent need for interventions to ensure equitable access to colorectal cancer treatment for patients with SSD, even in the context of a universal healthcare system.
- Limitations
  - Survival outcomes were not available; therefore, the impact of treatment disparities on mortality could not be assessed.
  - As the reasons for not receiving treatment were unknown, it was not possible to distinguish between medically justified decisions, patient preferences, and disparities in access to care.

at institutions participating in the Quality-of-care Monitoring Project. Colorectal cancer cases were identified using ICD-O-3 codes, as detailed in Table S1. Patients with SSD were defined as those whose DPC inpatient records included comorbidities coded F20–F29 according to the International Classification of Diseases, Tenth Revision (ICD-10), within 1 year following colorectal cancer diagnosis. Patients were defined as having no mental disorder if no ICD-10 F codes were listed among their comorbidities at the time of admission.

Patients who did not receive inpatient treatment within 1 year of diagnosis at the facility where they were treated for colorectal cancer were excluded from the study, as psychiatric comorbidity data are available only for hospitalized cases. Additional exclusion criteria were as follows: age under 18 years at the time of diagnosis; receipt of neoadjuvant chemotherapy (as this treatment alters the standard therapeutic sequence); psychiatric diagnoses other than SSD; and missing data on the Barthel Index at the first hospitalization following cancer diagnosis.

2.5 | Outcomes

We evaluated two study outcomes. The first outcome was whether the patient underwent surgical or endoscopic treatment for colorectal cancer. These procedures were identified through DPC records using the original Japanese surgical procedure codes for colectomy, endoscopic mucosal resection, and endoscopic submucosal dissection. The second outcome was the receipt of stage-appropriate systemic therapy other than surgery: (i) adjuvant chemotherapy using standard regimens within 56 days postoperatively for patients with pathological stage III disease (see Table S2), and (ii) systemic therapy with standard regimens for patients with stage IV disease who did not undergo surgical treatment (see Table S2) [26, 27].

2.6 | Covariates

Based on previous studies [12–18], the following covariates were included in the analyses: age, sex, clinical stage, and scores on the Charlson Comorbidity Index and Barthel Index. The Barthel Index, which assesses functional independence in a similar way to performance status scales, has been validated as a prognostic factor in patients with colorectal cancer [28, 29].

2.7 | Statistical Analyses

Baseline characteristics were compared between patients with a diagnosis of SSD and those with no psychiatric diagnosis. Continuous variables were assessed using the Mann–Whitney U test and categorical variables were compared using chi-square tests.

To evaluate whether SSD status was independently associated with the receipt of surgical or endoscopic treatment, we conducted multivariable logistic regression, adjusting for age, sex, clinical stage, and scores on the Charlson Comorbidity Index and Barthel Index. We also conducted analyses stratified by sex and by cancer site (colon versus rectum). Similar multivariable logistic regression models were used to evaluate the association between SSD status and receipt of stage-appropriate systemic therapy.

All statistical tests were two-sided, and analyses were performed using StataMP 17.0 (StataCorp LP, College Station, TX, USA). p-values < 0.05 were considered statistically significant.

3 | Results

Figure 1 illustrates the flow chart for patient inclusion. We included 248,966 patients from 709 hospitals, of whom 2337 were identified as having SSD.

Table 1 presents the patients' baseline characteristics. Among those with no psychiatric disorder, 10.0% were diagnosed at stage 0 and 20.5% at stage 1, whereas the corresponding proportions for those with SSD were 5.4% and 15.2%, respectively.

Table 2 and Table S3 display the results of the multivariable logistic regression analysis regarding the odds of receiving surgical or endoscopic treatment. Patients with SSD were significantly less likely to receive surgical or endoscopic treatment for colorectal cancer than those without psychiatric disorders (adjusted odds ratio [aOR], 0.83; 95% confidence interval [CI], 0.73–0.94; $p = 0.004$). In the stratified analysis by sex, a significant association was observed among men (aOR, 0.79; 95% CI, 0.66–0.94; $p = 0.007$), but not among women (aOR, 0.90; 95% CI, 0.75–1.07; $p = 0.23$). In the stratified analysis by cancer site, a significant association was observed for colon cancer (aOR, 0.82; 95% CI, 0.70–0.95; $p = 0.01$), but not for rectal cancer (aOR, 0.84; 95% CI, 0.68–1.05; $p = 0.13$).

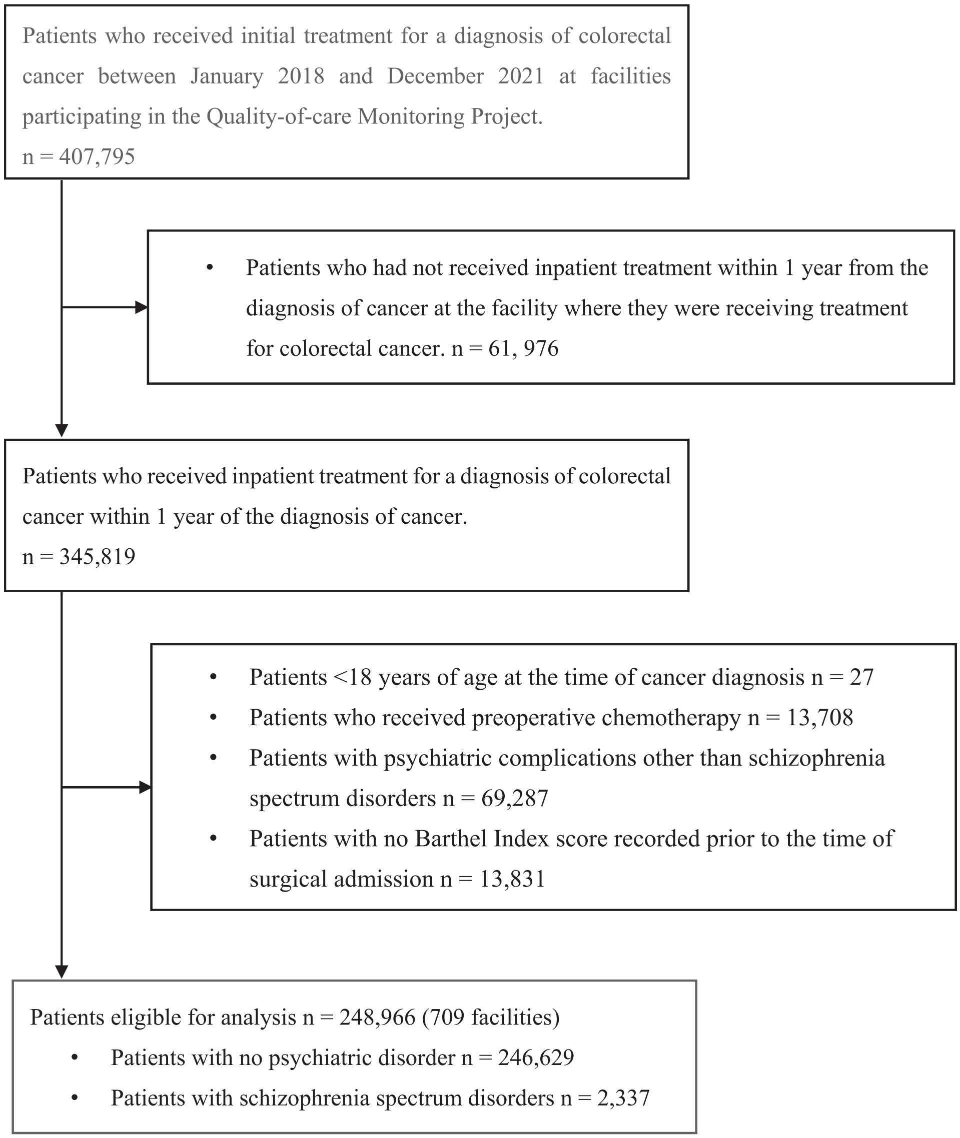

FIGURE 1 | Flow chart for patient inclusion.

Table 3 presents the results of the multivariable logistic regression analyses for the receipt of stage-appropriate systemic therapy. Patients with SSD were significantly less likely to receive adjuvant chemotherapy using standard regimens within 56 days after surgery for pathological stage III colorectal cancer (aOR, 0.33; 95% CI, 0.26–0.41; $p < 0.001$). For stage IV disease, patients with SSD were also less likely to receive systemic therapy with standard regimens (aOR, 0.23; 95% CI, 0.17–0.31; $p < 0.001$). Table S4 presents the patient characteristics in the analysis of adjuvant chemotherapy using standard regimens for patients with pathological stage III disease. Table S5 presents the patient characteristics in the analysis of systemic therapy for patients with stage IV disease who did not undergo surgical treatment.

4 | Discussion

This result of this study demonstrate that in Japan, patients with SSD were significantly less likely to receive surgery or endoscopic treatment for colorectal cancer than those without psychiatric disorders. Furthermore, disparities in stage-appropriate systemic therapy were particularly pronounced. For both postoperative adjuvant chemotherapy in stage III disease and systemic therapy in stage IV disease, the odds of patients with SSD receiving treatment were less than one-third that of patients without psychiatric disorders. A key strength of this study lies in its use of a nationwide dataset and its evaluation of stage-appropriate systemic therapy, as defined by Japanese colorectal cancer treatment guidelines.

Regarding clinical characteristics, patients with SSD were less likely to be diagnosed at an early clinical stage than those without psychiatric disorders. Previous studies have shown mixed findings: some reported no significant difference in colorectal cancer stage at diagnosis between patients with and without SMI [17, 30], whereas others found that patients with SMI were more likely to be diagnosed at later stages [12, 15, 31]. In Japan, individuals aged 40 and older are offered annual population-based colorectal cancer screening using fecal occult blood testing [32]. Previous studies have shown that participation in colorectal cancer screening is substantially lower among patients with SSD

TABLE 1 | Patient Characteristics.

	All patients (*N* = 248,966)	Patients without psychiatric disorders (*n* = 246,629)	Patients with schizophrenia spectrum disorders (*n* = 2337)	*p*
Age in years, median (IQR)	72 (65–79)	72 (65–79)	71 (63–80)	0.114
Sex (female), *N* (%)	103,265 (41.5)	102,125 (41.4)	1140 (48.8)	< 0.001
Clinical stage of colorectal cancer, *N* (%)				
0	24,723 (9.9)	24,598 (10.0)	125 (5.4)	< 0.001
I	50,789 (20.4)	50,434 (20.5)	355 (15.2)	
II	44,604 (17.9)	44,066 (17.9)	538 (23.0)	
III	57,681 (23.2)	57,087 (23.2)	594 (25.4)	
IV	40,778 (16.4)	40,281 (16.3)	497 (21.3)	
Unknown	30,391 (12.2)	30,163 (12.2)	228 (9.8)	
Charlson Comorbidity index, median (IQR)	3 (2–4)	3 (2–4)	3 (2–6)	< 0.001
Charlson Comorbidity index, *N* (%)				
2	120,099 (48.2)	119,167 (48.3)	932 (39.9)	< 0.001
3	55,523 (22.3)	54,968 (22.3)	555 (23.8)	
4	18,707 (7.5)	18,536 (7.5)	171 (7.3)	
5	5880 (2.4)	5808 (2.4)	72 (3.1)	
≥ 6	48,757 (19.6)	48,150 (19.5)	607 (25.9)	
Barthel index, *N* (%)				
100	211,220 (84.8)	209,936 (85.1)	1284 (54.9)	< 0.001
60–95	18,243 (7.3)	17,908 (7.3)	335 (14.3)	
≤ 55	19,503 (7.8)	18,785 (7.6)	718 (30.7)	
Cancer site, *N* (%)				
Colon	171,237 (68.8)	169,625 (68.8)	1612 (69.0)	0.84
Rectum	77,729 (31.2)	77,004 (31.2)	725 (31.0)	
Patients undergoing surgical or endoscopic treatment, *N* (%)	219,627 (88.2)	217,788 (88.3)	1839 (78.7)	< 0.001

Abbreviation: IQR, interquartile range.

in Japan [33–35], which may partly explain the higher proportion diagnosed at advanced stages.

This study found that patients with SSD were significantly less likely to receive invasive treatments for colorectal cancer than those without psychiatric disorders. A Japanese study by Ishikawa et al., using the DPC database, reported that patients hospitalized for gastrointestinal cancer in 2012 who had comorbid schizophrenia were significantly less likely to receive invasive treatments [36]. Although a direct comparison with the study by Ishikawa et al. is not possible because our study focused specifically on colorectal cancer, our findings confirm that considerable disparities in access to invasive treatments remain. Similar disparities have been reported in other countries with universal health insurance systems [13, 16, 17], although a recent study in Sweden found no significant difference in the receipt of surgical resection for stage I–III colorectal cancer between patients with and without SMI [18]. Osterman et al. suggested that in Sweden, disparities may have been reduced not only as a result of access to universal healthcare but also because of the introduction of standardized treatment pathways and mandatory multidisciplinary team conferences [18]. These findings suggest that reducing disparities in invasive treatments for patients with SSD requires tackling not only financial barriers but also other barriers to care.

The sex-stratified analyses showed that female patients with SSD were as likely as those without psychiatric disorders to receive invasive treatments, whereas male patients with SSD were significantly less likely to receive such treatments. To the best of our knowledge, no previous studies have investigated sex differences in disparities in invasive colorectal cancer

TABLE 2 | Odds of receiving surgical or endoscopic treatment based on the presence of a schizophrenia spectrum disorder diagnosis.

	Total number/number of patients who received surgery	Crude odds ratio (95% CI)	*p*	Adjusted odds ratio (95% CI)	*p*
Overall (*N* = 248,966)					
Without psychiatric disorders	246,629/217,788	Ref		Ref	
With schizophrenia spectrum disorders	2337/1839	0.49 (0.44–0.54)	< 0.001	0.83 (0.73–0.94)[a]	0.004
Stratified by sex					
Men (*N* = 145,701)					
Without psychiatric disorders	144,504/127,927	Ref		Ref	
With schizophrenia spectrum disorders	1197/938	0.47 (0.41–0.54)	< 0.001	0.79 (0.66–0.94)[b]	0.007
Women (*N* = 103,265)					
Without psychiatric disorders	102,125/89,861	Ref		Ref	
With schizophrenia spectrum disorders	1140/901	0.51 (0.45–0.59)	< 0.001	0.90 (0.75–1.07)[b]	0.23
Stratified by cancer site					
Colon (*N* = 171,237)					
Without psychiatric disorder	169,625/151,540	Ref		Ref	
With schizophrenia spectrum disorder	1612/1284	0.47 (0.41–0.53)	< 0.001	0.82 (0.70–0.95)[a]	0.01
Rectum (*N* = 77,729)					
Without psychiatric disorders	77,004/66,248	Ref		Ref	
With schizophrenia spectrum disorders	725/555	0.53 (0.45–0.63)	< 0.001	0.84 (0.68–1.05)[a]	0.13

Note: Sex-stratified analyses are presented as adjusted odds ratios only.
Abbreviation: CI, confidence interval.
[a]Adjusted for age category, sex, clinical stage of colorectal cancer, Charlson Comorbidity Index score, and Barthel Index score.
[b]Adjusted for age category, clinical stage of colorectal cancer, Charlson Comorbidity Index score, and Barthel Index score.

treatment among patients with SSD. These findings might partly reflect that male patients with SSD report more severe symptoms and poorer functioning compared with female patients [37, 38].

Regarding the stratified analysis by cancer site, the point estimates for the odds of receiving invasive treatment were comparable between colon (aOR, 0.82) and rectal cancer (aOR, 0.84). The association did not reach statistical significance for rectal cancer, which may be partly explained by the smaller sample size of this subgroup. Previous studies have shown that patients with SMI are more likely to undergo emergency surgery [15, 18, 31] and less likely to receive sphincter-preserving surgery [14] than those without psychiatric disorders. Therefore, further investigation is needed to clarify potential disparities in the surgical approach or specific procedures performed.

The present study revealed that patients with SSD were substantially less likely to receive guideline-recommended postoperative chemotherapy compared with patients without psychiatric disorders. Disparities in the receipt of adjuvant chemotherapy among patients with SMI have also been reported in Canada [16], France [17], and Sweden [18]. Our findings further demonstrate that SSD is associated with a lower likelihood of receiving postoperative chemotherapy in Japan. Because postoperative chemotherapy is typically delivered in outpatient settings, adherence to treatment and self-care for managing adverse effects may pose particular challenges for patients with SSD [8–10, 39].

Furthermore, treatment disparities were more pronounced among patients with stage IV disease. Those with SSD were significantly less likely to receive guideline-recommended

TABLE 3 | Odds of receiving stage-appropriate treatment for colorectal cancer based on presence of a schizophrenia spectrum disorder diagnosis.

	Total number/number of patients who received treatment	Crude odds ratio (95% CI)	p	Adjusted odds ratio (95% CI)	p
Postoperative chemotherapy with standard regimens in patients with pathological stage III (n = 50,720)					
Without psychiatric disorders	50,214/28,496	Ref		Ref	
With schizophrenia spectrum disorders	506/133	0.27 (0.22–0.33)	< 0.001	0.33 (0.26–0.41)[a]	< 0.001
Systemic therapy with standard regimens in patients with stage IV who did not undergo surgical treatment (n = 21,005)					
Without psychiatric disorders	20,705/12,603	Ref		Ref	
With schizophrenia spectrum disorders	300/83	0.25 (0.19–0.32)	< 0.001	0.23 (0.17–0.31)[a]	< 0.001

Abbreviation: CI, confidence interval.
[a]Adjusted for age category, sex, Charlson Comorbidity Index score, and Barthel Index score.

systemic therapy than patients without psychiatric disorders. Consistent with our findings, Seppänen et al. reported that patients with metastatic colorectal cancer and comorbid SMI were less likely to receive chemotherapy [17]. Treatment options for stage IV disease are diverse, and systemic therapy is typically administered not with curative intent, but based on the patient's condition, preferences, and the physician's clinical judgment. This complex decision-making process may inadvertently disadvantage patients with SSD, who often face challenges in communication and shared decision-making. Our finding that patients with SSD are more likely to be diagnosed at advanced stages makes this disparity particularly concerning. Given that recent advances in systemic therapy have significantly improved survival outcomes [20], this gap in access to effective treatment may contribute to the higher mortality rates among individuals with SSD.

Taken together, our findings illustrate a concerning reality in which patients with SSD face compounding disadvantages across the cancer care continuum. This highlights the urgent need for a two-pronged approach to mitigate these disparities. First, to address the problem of late-stage diagnosis, interventions to improve screening uptake are essential. In Japan, patient navigation and educational programs have already proven effective in increasing colorectal cancer screening participation among this population [40]. Second, to close the profound gaps in cancer treatment, particularly in systemic therapy, it is essential to develop systems that support patients with SSD. Integrated support involving both cancer care and mental health providers, such as collaborative care models, could offer a promising framework [41, 42].

This study has several limitations. First, survival outcomes were not available; therefore, it remains unclear whether the observed disparities in colorectal cancer treatment among patients with SSD are associated with differences in mortality. Second, because this was a database study, the reasons for not receiving cancer treatment were unknown; we were therefore unable to distinguish between cases where treatment was withheld due to valid medical reasons or patient preferences. Third, the study population excluded patients who were not hospitalized at participating cancer care institutions after cancer diagnosis. Patients with more severe psychiatric symptoms may be less likely to be admitted to cancer hospitals, potentially leading to an underestimation of disparities. Fourth, we were unable to assess whether patients completed the entire course of stage-appropriate systemic therapy. Fifth, patients with mild or remitted SSD symptoms may not have had SSD recorded as a comorbidity in the DPC database, leading to preferential identification of more severe cases. Consequently, the impact of SSD on cancer treatment might have been overestimated. Sixth, we were unable to retrieve detailed clinical information on SSD, including the severity of psychiatric symptoms and specific medications (e.g., antipsychotics). These unmeasured factors may limit the generalizability of our findings and may have influenced treatment choices.

5 | Conclusion

This study demonstrated that even in Japan—a country with a universal health insurance system—patients with SSD face significant disparities in colorectal cancer treatments. While a gap was evident in the receipt of invasive treatments, the disparities were greater for stage-appropriate systemic therapy, which is a critical component of care across different disease stages. These findings underscore the need for interventions that not only address disparities in surgery but also ensure equitable access to systemic therapy and other stage-specific treatments throughout the cancer care continuum for this vulnerable population.

Author Contributions

Conceptualization: all authors. Data curation: Taisuke Ishii and Tomone Watanabe. Formal analysis: Taisuke Ishii and Tomone Watanabe. Funding acquisition: Masaki Fujiwara and Inagaki M. Investigation: Masaki Fujiwara, Yuto Yamada, and Taisuke Ishii. Methodology: all authors. Project administration: Masaki Fujiwara, Yuto Yamada, and

Masatoshi Inagaki. Supervision: Masatoshi Inagaki. Visualization: Masaki Fujiwara and Yuto Yamada. Writing original draft: Masaki Fujiwara and Yuto Yamada. Review and editing: all authors.

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Ethics Statement

This study was approved by the ethics committee of Okayama University (KEN2407-001) and was conducted in accordance with the Declaration of Helsinki. Information regarding this study, including patients' and their families' right to opt out of participation, has been disclosed on the website of the National Cancer Center Japan.

Conflicts of Interest

Fujiwara M. has received honoraria from Eisai. Yamada Y. has received honoraria from Otsuka, Daiichi Sankyo, Sumitomo Pharma, Shionogi, Takeda, and Lundbeck. Otsuki K. has received honoraria for lectures from Otsuka, Meiji, Sumitomo Pharma, Viatris, and Takeda. Uchitomi Y. has received grants from Daiichi Sankyo, Aflac Life Insurance Japan, SUSMED, Meiji, and Shionogi. Inagaki M. has received grants from Astellas, Eisai, Otsuka, Shionogi, Daiichi Sankyo, Sumitomo Pharma, Takeda, Mitsubishi Tanabe Pharma, Nihon Medi-Physics, Pfizer, Fujifilm, and Mochida and has received honoraria for lectures from EA Pharma, Meiji, MSD, Viatris, Eisai, Otsuka, Sumitomo Pharma, Takeda, Eli Lilly, Nippon Shinyaku, Pfizer, Mochida, Janssen, and Yoshitomi Yakuhin. The other authors declare no conflicts of interest.

Data Availability Statement

Special permission to use the data was granted by the privacy protection law and data use agreement with the hospitals, and the data are not publicly available.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.